R1 Forces and Newton's laws

Name: ______________________

Class: ______________________

Date: ______________________

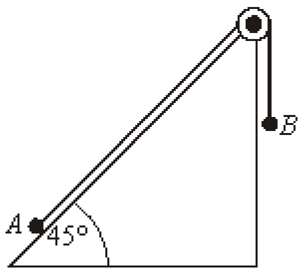
Time: **55 minutes**

Marks: **45 marks**

Comments:

Q1.

Two particles, A and B, are connected by a light inextensible string, which passes over a smooth light pulley. Particle A is on a smooth slope, at 45° to the horizontal, and particle B hangs with the string vertical, as shown in the diagram.



The mass of A is 14 kg and the mass of B is 6 kg.

(a) Using two equations of motion, show that the acceleration of the particles is 1.91 m s^{-2} correct to three significant figures.

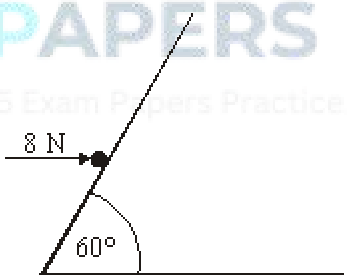
(6)

(b) Particle B is replaced by a particle C of mass m kg. After the particles have been set in motion, they move with a constant speed. Find m.

(4)
(Total 10 marks)

Q2.

A particle is held at rest on a smooth slope by a horizontal force of magnitude 8 newtons, as shown in the diagram below. The slope is at an angle of 60° to the horizontal.



(a) Draw a diagram to show the forces acting on the particle.

(1)

(b) By resolving horizontally show that the magnitude of the normal reaction force acting on the particle is approximately 9.24 newtons.

(3)

(c) Find the mass of the particle, giving your answer to two significant figures.

(3)
(Total 7 marks)

Q3.

A box, of mass 20 kg, is initially at rest on a rough horizontal surface. A horizontal force of magnitude P newtons is applied to the box. The coefficient of friction between the box and the surface is 0.3.

(a) State the magnitude of the normal reaction force acting on the box.

(1)

(b) Find the magnitude of the friction force that acts on the box if:

(i) $P = 80$;

(ii) $P = 40$.

(3)

(c) Find the value of P when the box is accelerating at 0.8 m s^{-2}.

(3)

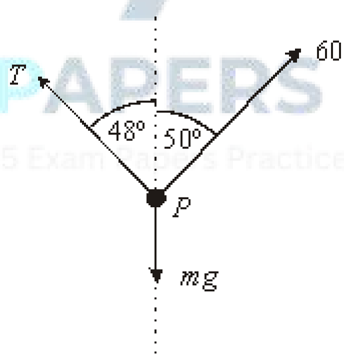
(d) When the box reaches a speed of 6 m s^{-1}, the horizontal force P is removed. Find the distance that the box travels after the force P is removed.

(5)

(Total 12 marks)

Q4.

A particle, P, of mass m kg, is held in equilibrium by two strings. One string is inclined at 50° to the vertical and exerts a force of 60 newtons on the particle. The other string exerts a force of magnitude T newtons at an angle of 48° to the vertical. The forces that act on the particle are shown in the diagram below.



(a) Find T.

(3)

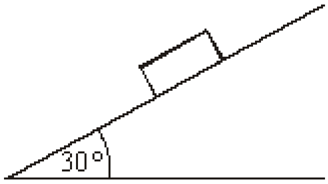
(b) Find m.

(4)

(Total 7 marks)

Q5.

A block, of mass 7 kg, is placed on a rough slope that is inclined at 30° to the horizontal, as shown in the diagram. The block remains at rest in this position.



(a) Draw a diagram to show the forces acting on the block.

(1)

(b) Find the magnitude of the normal reaction force acting on the block.

(2)

(c) Find the magnitude of the friction force acting on the block.

(2)

(d) The coefficient of friction between the block and the plane is μ. Find an inequality that μ must satisfy.

(2)

(e) A similar block, of mass 14 kg, is placed on the slope. Does this block remain at rest or slide? Give a reason for your answer.

(2)

(Total 9 marks)